

Contents

Bricks ISPF Editor — User Manual	1
Contents	1
Access	2
Menu	2
File browser	3
Browser command line	3
Browser PF keys	4
Filename rules (PF6 “new file”)	4
Editor screen layout	4
PF keys	5
Command-line commands	6
Find / change	6
Navigation	6
Save and exit	7
Display toggles	7
Other	7
Line-prefix commands	8
Editing the buffer	8
Copy / move / paste	8
Case and indent	8
Exclusion	8
Inserting at file boundaries	9
Block commands	9
Save flow and syntax validation	9
Multi-file editing	10
Opening a second (third, ...) file	10
Cycling between buffers	10
Closing a buffer — the hard rule	10
Edit locks across buffers	11
The 9-buffer limit	11
What survives a F2 round-trip	11
Edit locks	11
Syntax highlighting	12
Limitations	12

Bricks ISPF Editor — User Manual

The ISPF transaction is a built-in source editor that lets operators browse and edit the REXX, COBOL, and BMS-map source trees from a 3270 session. It models the classic IBM ISPF editor: a command line at the top, a 6-byte prefix area on every content row for line commands, and PF-key shortcuts.

Contents

- Access

- Menu
 - File browser
 - Editor screen layout
 - PF keys
 - Command-line commands
 - Line-prefix commands
 - Block commands
 - Save flow and syntax validation
 - Multi-file editing
 - Edit locks
 - Syntax highlighting
 - Limitations
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Access

ISPF is a built-in TRANSID. It does **not** appear in `runtime/transactions.conf`; it's dispatched by name from the bricks core. To use it you must:

1. Sign on with CSSN (the bricks logon transaction).
2. Belong to the dev group in `runtime/users.conf`.

Type ISPF at the blank prompt and press ENTER. Non-dev users see ISPF requires DEV group membership. and are returned to the prompt.

Menu

The menu screen offers three editing areas:

```

----- ISPF EDIT UTILITY -----
OPTION  ==>  _

      1 - Edit REXX programs      (runtime/rexx)
      2 - Edit COBOL programs    (runtime/cobol)
      3 - Edit BMS map files     (runtime/map)

```

Press ENTER after typing the option number, or F3 to exit ISPF.

...

F3=Exit

Type 1, 2, or 3 in the OPTION ==> field and press ENTER. PF3 exits ISPF and returns you to the blank prompt.

The directories are taken from your `bricks.cnf`:

Area	Config key	Default
REXX	rexx_dir	runtime/rexx

Area	Config key	Default
COBOL	cobol_dir	runtime/cobol
MAPS	maps_dir	runtime/map

File browser

After picking an area, the browser shows the directory contents in a two-column paged list:

```

----- FILE BROWSER -- REXX -- Sort: NAME -- Page 1/1
Command ==>                               Sort: N=Name D=Date
S Filename      Size Date  Syntax      S Filename      ...
  cons.rexx      3787 01May  OK
  cusl.rexx      4458 01May  Err:127
  cust.rexx      7167 01May  OK
...
F3=Menu  F6=New  F7=Up  F8=Down  any char=Open  D=Delete

```

Each row has a single-byte **selector field** (the S column) to its left. Type a character into the selector and press ENTER:

Selector	Action
Any letter except D	Open the file in the editor
D	Delete the file (a confirmation overlay appears; press F9 to confirm, anything else cancels)

The right-most **Syntax** column shows the live parse status of every file in the directory:

Value	Meaning
OK (green)	The file parses cleanly with the area's parser.
Err:N (red)	The parser reported a syntax error at line N.
Err (red, no line)	I/O error or no line number available.

Browser command line

Type into the Command ==> field:

Command	Effect
N or NAME	Sort by filename (case-insensitive).
D or DATE	Sort by modification time, newest first.

Browser PF keys

Key	Action
F3	Return to the menu.
F6	Create a new file. Prompts for a filename, auto-appends the area extension (. rexx, . cob, . map) if you omit it, validates the name, creates an empty file, acquires the edit lock, and drops into the editor.
F7	Page up.
F8	Page down.
F9	(Only when the delete-confirmation overlay is showing.) Confirm the delete.

Filename rules (PF6 “new file”)

The browser’s new-file prompt enforces a sandbox:

- Non-empty.
- No /, \, NUL byte, or .. substring.
- ASCII printable bytes only (0x20–0x7E).
- Must not already exist on disk.
- The area extension is auto-appended if the name has no ..

A path that resolves outside the area directory is rejected.

Editor screen layout

```
EDIT  runtime/rexx/cusl.rexx          Columns 00001 00071  1
Command ==> _                        Scroll ==> PAGE_
***** ***** Top of Data *****
' ' ' ' ' /* CUSL -- list / search the customers file, ... */
' ' ' ' ' ADDRESS CICS
...
***** ***** Bottom of Data *****
```

- **Row 0** — EDIT <title> plus Columns NNNNN NNNNN showing the visible-column range. The digit at column 78 is the active-file number (1 to 9) when multiple files are open. Error and status messages also paint here, after column 23.
- **Row 1** — Command ==> writable field, then Scroll ==> writable field showing the current scroll mode.
- **Row 2 (only at top of file)** — Top of Data marker. The ***** segment at the start of this row is writable: type a line command (e.g. I3) here to insert lines before line 1.
- **Content rows** — each has a 6-byte prefix (' ' ' ' ' ' by default, or 000123 when NUMBER ON is active, or CC etc. when a block command is pending), then the line content.
- **Bottom-of-Data row** — Bottom of Data marker. Its ***** segment is writable too: type I3 here to append blank lines past the last content line.

Prefix colors:

Color	Meaning
Red	Default — unchanged line.
Yellow	Cursor's current row.
Turquoise	Line modified in the most recent interaction.

PF keys

Key	Action
F1	Show the help overlay (any key dismisses).
F2	Open the file browser without closing the editor. The current file stays open in the multi-file slot; pick another file to open it as buffer 2.
F3	Close the current buffer. If it's modified, an abandon-confirmation overlay appears; F9 confirms the discard. The editor stays alive as long as any other buffer is open — it switches to the next remaining buffer. Only when the last buffer is closed does the editor return to the browser. See Multi-file editing.
F4	Open this manual's help text as an additional editable buffer. F9 cycles back to your file.
F5	Repeat the last FIND (continues from the previous match).
F7	Scroll up by the amount in the Scroll ==> field.
F8	Scroll down by the amount in the Scroll ==> field.
F9	Cycle to the next open file in the multi-file slot.
F10	Scroll left by 8 columns.
F11	Scroll right by 8 columns.
F12	Save the current buffer and close it (same close-and-cycle behaviour as F3 on a clean buffer). On a parse error, the message paints on row 0 and the buffer stays open; press F12 again to save anyway.

The Scroll ==> field accepts:

Mode	Lines per F7 / F8
CSR (default)	1

Mode	Lines per F7 / F8
HALF	half the visible-line count
PAGE	full visible-line count
DATA	visible-line count minus 1 (always leaves one row of context)

Partial input is resolved by prefix and Levenshtein distance: P □ PAGE, PGE □ PAGE, CURSOR □ CSR.

Command-line commands

Type into the Command ==> field and press ENTER. Verb matching is case-insensitive; quoted arguments preserve case.

Find / change

Command	Effect
FIND text or F text	Locate the next occurrence of text (case-insensitive). Wraps from end of file back to the start. The cursor lands on the match, every match across the buffer is highlighted in red + underscore, and the status line shows Found N matches.
F (no argument) or RFIND	Repeat the last FIND from the current cursor position. Same as PF5.
CHANGE old new or C old new	Replace the first occurrence of old with new and report Changed N. The status line shows not found. if the search string isn't present.
CHANGE old new ALL or C old new ALL	Replace every occurrence in every line.

Quotes can be used to embed spaces: FIND 'no records' / CHANGE 'a b' 'c d'.

Navigation

Command	Effect
LOCATE n / LOC n / L n	Set the top-line cursor to line N (1-indexed). Clamped to the valid range.
TOP / T	Scroll to the start of the buffer.
BOTTOM / BOT	Scroll to the end.

Save and exit

Command	Effect
SAVE	Persist the current buffer to disk and stay in the editor (does not close the buffer — unlike F12). On a parse error the message is shown on row 0; a second SAVE (or F12) saves anyway.
CANCEL / CAN	Close the current buffer without saving, no abandon-confirmation prompt. Same close-and-cycle behaviour as F3-then-F9: if other buffers are open, the editor switches to the next one; only the last buffer triggers an exit.
UNDO	Restore the buffer from the most recent undo snapshot. Same as the U line command from the prefix area.

Display toggles

Command	Effect
NUMBER ON / NUM ON	Show 6-digit line numbers in the prefix area instead of ' ' ' ' ' ' ' '.
NUMBER OFF / NUM OFF	Restore the ' ' ' ' ' ' ' ' placeholder.
NUMBER / NUM	Toggle the current setting.
COLS / COL	Toggle the column ruler at row 2 (replaces the Top-of-Data marker while on).
HI ON	Turn syntax highlighting on.
HI OFF	Turn syntax highlighting off.
HI	Toggle the current setting.

Other

Command	Effect
RESET / RES	Clear FIND highlights, the column ruler, excluded line ranges, and the last-command echo. Does not clear the turquoise “recently modified” prefix markers.
HELP	Show the help overlay. Same as F1.
EDIT	Open the file browser without leaving the editor. Same as F2.

Anything else lands on row 0 as Invalid command.

Line-prefix commands

Type into the 6-byte prefix area on the line you want to act on and press ENTER. Whitespace and leftover ' ' characters are stripped before parsing. Most verbs accept an optional numeric count (D3 = delete 3 lines, R5 = repeat 5 times).

Editing the buffer

Command	Effect
D / Dn	Delete this line / N lines starting here.
I / In	Insert N blank lines (default 1) after this line.
R / Rn	Repeat this line N times immediately after itself.

Copy / move / paste

Command	Effect
C / Cn	Copy this line / N lines into the copy buffer. The lines stay where they are.
M / Mn	Move this line / N lines into the move buffer (the source lines are removed).
A	Paste the most recent copy/move buffer after this line.
B	Paste before this line.
O	Overlay — paste the copy/move buffer onto this line and the lines below it, replacing their content row-for-row.

The copy/move buffer is cleared after A, B, or O consumes it. The TOD and BOD markers also accept A (after) and B (before) to paste at the buffer boundaries.

Case and indent

Command	Effect
U	Uppercase this line.
L	Lowercase this line.
) /)n	Shift this line right N spaces (default 1).
(/ (n	Shift this line left N spaces (up to the first non-space character).

Exclusion

Command	Effect
X / Xn	Exclude this line / N lines from view. Excluded lines collapse into a turquoise summary row: – – 5 Line(s) excluded. Type RESET to bring them back.

Inserting at file boundaries

The ***** segment of the Top-of-Data and Bottom-of-Data marker rows is writable. Type an I or In there to insert blank lines before line 1 (TOD) or after the last line (BOD).

Block commands

Doubled-verb forms mark a range. Type the verb on the first line of the range, press ENTER (the prefix shows the pending verb, e.g. CC), then on a later screen type the same doubled verb on the last line of the range and press ENTER. The block executes:

Form	Single equivalent	Effect on the range
DD..DD	D	Delete every line.
CC..CC	C	Copy every line into the copy buffer.
MM..MM	M	Move every line into the move buffer (deletes the originals).
UU..UU	U	Uppercase every line.
LL..LL	L	Lowercase every line.
))..))	)	Shift the block right (count from either marker).
((..((	(	Shift the block left.
XX..XX	X	Exclude the block from view.

If you change your mind, type RESET on the command line to discard the pending half-block marker.

Save flow and syntax validation

Bricks runs the area's parser (rexx.Parse, cobol.Parse, mapdsl.Parse) on the buffer before writing. The flow is **warn-then- save**:

1. First PF12 or SAVE with a parser error: the error message is painted on row 0 in red and the buffer is **not** written. A bypass flag is armed internally.
2. Second PF12 or SAVE: the bypass is honored, the buffer is written regardless of the parse error.

This applies to every area (REXX, COBOL, MAP). A clean parse saves immediately on the first press.

The status messages match the canonical ISPF wording:

Message	When
Saved	Successful save (via PF12 or SAVE command).
Save error	The filesystem write failed.
<raw parser error>	First save with a parse error — bypass armed.

Multi-file editing

The editor holds **up to 9 buffers open simultaneously**. Each buffer keeps its own content, cursor position, scroll offset, undo snapshot, modified-line markers, and dirty bit. They never share state — editing one buffer can't affect another.

Opening a second (third, ...) file

Buffers are added to the slot when:

- The browser opens a file for the first time (always becomes buffer 1 if none are open).
- You press **F2** from inside the editor: the current buffer stays resident, the browser appears, you pick another file ☐ it opens as the next available buffer.
- You press **F4** to open this manual's help text as an additional editable buffer.

The current buffer's **file number** (1–9) is painted in yellow intense at column 78 of row 0, so you always know which slot you're on.

Cycling between buffers

F9 cycles forward through OpenFiles. Modified buffers keep their dirty bit on the round trip — turquoise prefix markers, the Modified flag, and any pending block-command markers all survive when you leave a buffer and come back.

Closing a buffer — the hard rule

The editor **never exits while other buffers are still open**. F3, F12, the CANCEL command, and the abandon-confirm overlay (F9 over "Changes have not been saved!") all act on the **active buffer only**:

Action	What happens to the active buffer	What happens to OTHER buffers
F3 (clean buffer)	Dropped from the slot	Untouched — editor cycles to the next
F3 (dirty buffer) ☐ F9 confirm	Dropped, edits discarded	Untouched — editor cycles to the next

Action	What happens to the active buffer	What happens to OTHER buffers
F12 (successful save)	Saved to disk, then dropped	Untouched — editor cycles to the next
F12 (validator refused)	Stays open with error on row 0	Untouched
CANCEL command	Dropped, edits discarded	Untouched — editor cycles to the next
SAVE command	Saved to disk, stays open	Untouched — operator keeps editing

The editor returns to the browser **only when the last open buffer is closed**. To exit the whole ISPF session with multiple buffers open you have to F3 / F12 / CANCEL each one in turn — every dirty buffer gets its own abandon-confirm so unsaved edits in buffer N can't be silently lost when you F3 buffer 1.

Edit locks across buffers

Each open buffer holds its own exclusive edit lock for the duration of the editing session. When you F9-cycle from buffer A to buffer B, the editor releases the lock on A and acquires the lock on B before the first render of B. A concurrent dev operator can grab A's lock the moment you leave it; when you F9 back, if A is now held by someone else you'll see the Locked by USER123 message and return to the browser — A's buffer is dropped from your slot since you no longer own the file.

The 9-buffer limit

The 10th open attempt errors with Maximum files open (9) on the file browser's error line. Close one of the existing buffers (F3 or F12) and try again.

What survives a F2 round-trip

When you F2 back to the browser and then ENTER on the same file you're already editing, the editor reuses the existing buffer instead of re-reading from disk. If another tool has modified the file on disk in the meantime, the row-0 banner shows External changes – buffer kept so you know your in-memory copy is stale.

Edit locks

While a file is open in the editor it is locked process-wide, **per-buffer**. A second dev operator who tries to open the same file in their browser sees a red Locked by USER123 since HH:MM message and the file's syntax-status column shows edit in turquoise. Locks are also consulted by the file browser before any D (delete) on the file — a delete against a held file is refused with the same "Locked by" message.

A buffer's lock is held from the moment it becomes the active buffer (open, F2-pick-again, or F9-cycle-to) until the moment it stops being active (F9 to another buffer, F3/F12/CANCEL closing it,

or the TCP connection dropping). Closing a buffer with F3 releases **only that buffer's** lock; other open buffers keep their locks until you close them too.

Locks release automatically when:

- You close the buffer with F3, F12, or CANCEL.
- You F9-cycle away from the buffer (and acquire the lock for the new active buffer).
- Your TCP connection drops or your sign-on times out.
- The bricks process panics during your session (deferred cleanup catches this).

The browser paints the **filename** of a currently-locked file in red intense so you can see contention without trying to open.

Syntax highlighting

Highlighting is **off by default** because per-line tokenizing has edge cases (notably REXX /* ... */ block comments that span lines). Turn it on with the HI command:

Command ==> HI ON

When on, the editor paints token-color overlays on each visible line:

Token class	Color
Keywords	Blue intense
Strings	Turquoise
Numbers	Yellow
Comments	Green
Identifiers / default	Green

The highlighter is per-language — picked from the area at editor open. The browser's per-file syntax-status indicator (OK / Err:N) uses the same parser independently of whether highlighting is on.

Limitations

- No regex search; FIND and CHANGE are plain-string, case-insensitive.
- Undo is single-level: each modifying line-command pass overwrites the previous snapshot. CHANGE does not capture undo (matches tsu).
- REXX /* ... */ block comments that span lines won't fully highlight past the opening line when HI ON is set.
- The 6-byte prefix area is the only place to type line commands; there is no menu-driven equivalent.
- File path validation in the browser rejects subdirectories under the area root — files only.